

SIM800L AT Commands Cheat Sheet

This cheat sheet summarizes the most useful AT commands for the SIM800L GSM/GPRS module. Each command can be sent via UART from a microcontroller or serial terminal. The module typically responds with 'OK' or an error code.

1. Basic AT Commands

Command	Description	Response
AT	Test communication	OK
ATE0	Turn echo off	OK
AT+CSQ	Check signal quality	+CSQ: <rss>,<ber>
AT+CCID	Get SIM card number (ICCID)	20-digit ICCID
AT+CREG?	Check network registration	+CREG: 0,1

2. SMS Commands

AT+CMGF=1	Set SMS text mode	OK
AT+CMGS="<number>"	Send SMS to given number	> (then text + CTRL+Z)
AT+CMGR=<index>	Read SMS at index	+CMGR: ...
AT+CMGD=<index>	Delete SMS	OK

3. Call Commands

ATD<number>;	Dial call	OK
ATH	Hang up call	OK
ATA	Answer incoming call	OK

4. GPRS and HTTP Commands

AT+SAPBR=3,1,"CONTYPE","GPRS"	Set bearer profile for GPRS	OK
AT+SAPBR=3,1,"APN","your.apn"	Set APN for SIM	OK
AT+SAPBR=1,1	Open GPRS context	OK
AT+HTTPINIT	Initialize HTTP service	OK
AT+HTTPPARA="URL","http://example.com"	Set URL for HTTP request	OK
AT+HTTPACTION=0	Perform HTTP GET	+HTTPACTION: 0,200,<data_length>
AT+HTTPREAD	Read HTTP response	Data output
AT+HTTPTERM	Terminate HTTP service	OK

Note: Commands should end with carriage return '\r' or '\r\n'. Use 9600 baud rate for most SIM800L modules.